

UrbanCart AI Customer Operations & Automation Solution

Final Requirements Analysis, Proposed Solution & Implementation Scope

Prepared for UrbanCart | September 2026

1. Executive Summary

UrbanCart's discovery requirements point to a fragmented customer-support and operations problem rather than a single chatbot problem. The proposed solution introduces an AI interaction layer on top of a structured operational data foundation and a trusted knowledge layer. It supports 24/7 chat and voice, routine support, lead qualification, basic order inquiry and controlled human escalation while preserving human judgment for complex cases.

2. Client Requirements

24/7 chat and phone support; accurate answers based on UrbanCart information; lead capture with name, phone, product, approximate budget, location and purchase readiness; basic existing-order inquiry; human escalation for damaged products, complex/disputed refunds, serious complaints, angry customers and other judgment-dependent cases; centralized customer/lead/order/support information; document-based RAG with automatic knowledge updates; selective Slack notifications; Notion internal documentation; reliability, security, easy maintenance, reasonable cost, scalability and an initial working version in approximately 6–8 weeks.

3. Current-State Problems

Routine product, price, availability, delivery and return questions are handled manually. After-hours inquiries can wait until the next business day. Customer information is fragmented across spreadsheets, the e-commerce system and complaint records. Order lookup is manual. Complex complaints need human judgment and immediate visibility. Product and policy information changes, while knowledge maintenance is manual. Slack should not become noisy with routine questions.

4. Proposed Solution

UrbanCart AI Customer Operations Hub: a 24/7 customer interaction layer backed by PostgreSQL/Supabase, Airtable operational records, Google Drive source documents, a Supabase vector/RAG knowledge layer, n8n orchestration, Notion internal documentation, Slack exception notifications, and a meaningful Zapier business automation for high-priority leads.

5. Technology Roles

Vapi — voice AI/receptionist. n8n — workflow orchestration and integrations. Zapier — business automation. Google Drive — authoritative document repository. Airtable — human-facing operational workspace. Notion — internal SOP/knowledge management. Slack — selective team notifications.

PostgreSQL/Supabase — structured relational data and backend/vector capabilities. RAG — grounded retrieval from UrbanCart knowledge documents.

6. Core Data Model

Customers; Products; Orders; Leads; Conversations; Messages; Support Tickets; Activities.

PostgreSQL/Supabase acts as the structured operational system of record, while Airtable provides a practical operational workspace for Sales, Support and management.

7. Customer & Team Journeys

Routine inquiry: customer → AI → PostgreSQL/RAG → grounded answer. Product inquiry: AI → Product Catalogue → confirmed information. Order inquiry: customer/order ID → Ordering Info → confirmed order data. Purchase journey: qualification → customer identification → lead creation → operational record → Sales notification when priority rules are met. Escalation: issue → human-intervention decision → support ticket → Slack notification → human support. Knowledge update: Google Drive document → n8n ingestion → extraction/chunking → embeddings → Supabase vector store → future grounded retrieval.

8. Automation vs Human Control

Automate routine product/policy questions, customer lookup, product lookup, basic order inquiry, lead capture, operational synchronization, RAG ingestion and selective notifications. Keep damaged products, serious/complex complaints, complex/disputed refunds, angry customers, explicit human requests, unsupported/unverified information and final resolution under human control.

9. RAG / Knowledge Architecture

UrbanCart product, shipping, return, warranty and support documents are stored in Google Drive. n8n retrieves and extracts content, prepares chunks and generates embeddings using OpenAI text-embedding-3-small. Supabase vector storage holds content, metadata and embeddings. The AI Agent invokes the RAG tool for UrbanCart-specific knowledge. Responses must be grounded in retrieved information; missing information must never be guessed. Updated source documents can be reprocessed without rewriting the AI prompt.

10. Human Escalation & Notifications

Qualifying cases include damaged products, serious/complex complaints, complex or disputed refunds, angry/highly escalated customers, explicit human requests, important unresolved order issues and cases requiring human judgment or unconfirmed information. The support ticket is the operational escalation record. After successful ticket creation, Slack receives a concise notification. Routine questions do not generate team-wide Slack alerts.

11. Implemented Demonstration Scope

Implemented: PostgreSQL/Supabase schema and fictional demo dataset; Airtable Leads, Support Tickets and Customers; n8n Supabase-to-Airtable synchronization workflows; Google Drive demo knowledge pack; RAG ingestion and retrieval; AI Agent tools for customer/product/order/lead/support/RAG operations; support

escalation and Slack notification; Zapier Airtable HIGH-priority lead → Sales Slack alert; Notion Internal Operations Hub and SOPs; Vapi voice-agent layer developed for the same underlying business logic.

12. Demonstration Scenarios

Routine policy question → grounded RAG answer. Product inquiry → structured product lookup. Existing order such as UC-10452 → confirmed order information. Purchase-intent lead → lead record plus HIGH-priority Sales Slack routing. Damaged product → support ticket plus human escalation. Unknown policy information → controlled human-assistance response. Voice call → Vapi receptionist using the same business logic.

13. Management Visibility

Airtable is the human-facing operational workspace. The recommended final presentation layer is an Airtable Interface dashboard showing lead volume and priority, open support tickets, customer activity and high-priority/escalated work. This dashboard complements rather than replaces PostgreSQL/Supabase, Notion, Google Drive or Slack.

14. Error Handling & Data Integrity

The AI must not claim a database write, ticket creation or notification succeeded unless the tool returns success. Missing or unverified business information results in human assistance rather than guessing. Order status, tracking, courier, delivery and payment details are reported only when returned by the transactional source. Escalation follows ticket creation before normal Slack notification. Automation failures are treated as failures and surfaced to operations.

15. Final Validation & Production Steps

Complete the Airtable Interface dashboard; run end-to-end regression tests for chat, voice, RAG, leads, order inquiry and escalation; test failure paths; review permissions and separation of internal/customer-facing knowledge; replace fictional demo data and demo knowledge with UrbanCart's authoritative production data/documents; finalize Vapi production configuration; publish validated workflows and document ownership/maintenance.

16. Important Demonstration Note

The discovery document states that UrbanCart has product, shipping, return, warranty and support documents, but it does not contain their actual contents. Therefore the demonstration uses clearly labelled fictional, internally consistent UrbanCart demo data and a demo knowledge pack. The RAG demonstration proves the mechanism; its policy content must not be represented as verified real-world UrbanCart policy until UrbanCart supplies authoritative documents.

17. Final Recommendation

Proceed as a phased customer-operations platform rather than attempting to automate every support process on day one. The proposed architecture directly addresses UrbanCart's stated priorities of reliability,

security, maintainability, reasonable cost, scalability and accurate AI responses, while preserving human control for complex cases and providing a clear path from demonstration to production.

Technology / Delivery Status Summary

Component	Purpose	Status
PostgreSQL/Supabase	Structured data + backend/vector	Implemented
Airtable	Operational records	Implemented
n8n	Orchestration/integration	Implemented
Google Drive + RAG	Knowledge ingestion/retrieval	Implemented
Notion	Internal SOPs/knowledge	Implemented
Slack	Human escalation + Sales alerts	Implemented/tested
Zapier	HIGH-priority lead → Sales Slack	Implemented/tested
Vapi	Voice AI layer	Built; final demonstration
Airtable Interface	Management cockpit	Final presentation layer